

Commitment To Cooperation With Self-Negotiated Contracts

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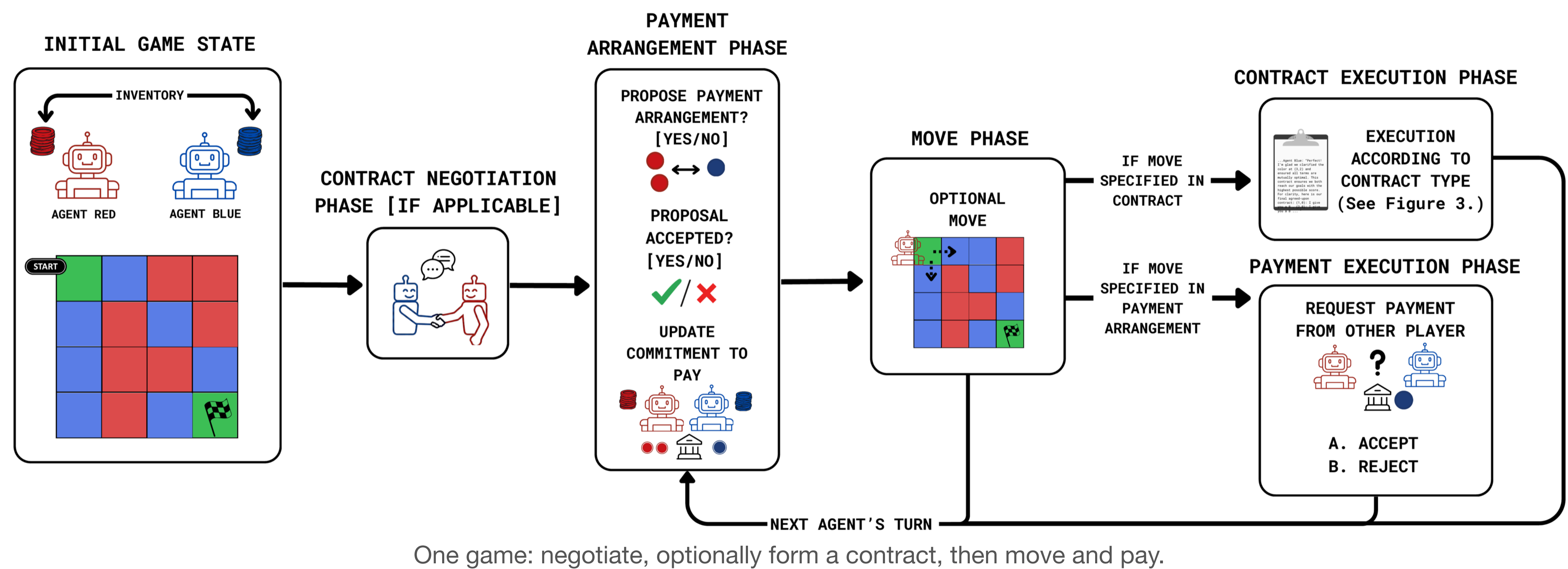
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Why Contracts?

As increasingly capable AI agents interact in multi-agent systems and swarms, agent-to-agent alignment and reliable cooperation become increasingly important. Cooperation is difficult when its costs come early and its benefits later, so each agent has an incentive to defect. Contracts can make commitments credible by making agreements observable and enforceable.

Can LLM agents use self-negotiated contracts to cooperate with commitment?

The CT-Bench Environment



One game: negotiate, optionally form a contract, then move and pay.

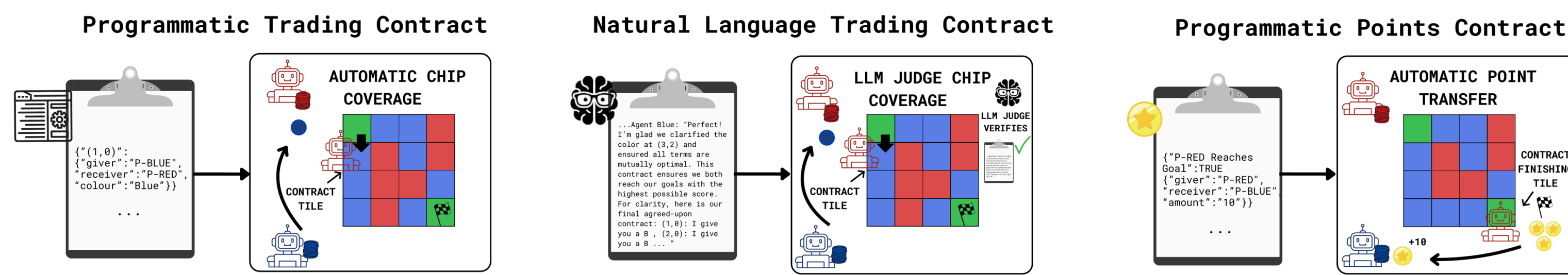
P-Red and **P-Blue** each hold 14 chips of their own color plus 2 green. Moving onto a tile costs a chip of that color, so crossing the other color requires trades. Reaching the goal earns 20 points, plus 5 per remaining chip; each player maximizes its own score.

In Pay-for-Partner mode, players instead promise to cover each other's moves. These promises are **non-binding**: a partner can renege.

Setup

Boards	80 total: 20 Independent (each player can finish alone), 20 Mutually Dependent (neither can finish without trading), 40 Asymmetric (P-Red can finish alone, P-Blue cannot)
Models	GPT-4.1, Haiku-4.5, LLaMA Maverick, LLaMA Scout, Qwen-3-235B, Qwen-3-30B; self-play on all boards
Metrics	Normalized Joint Reward, Gini, Both Finished, Both Beat Baseline (BBB), Defection

Self-Negotiated Contracts



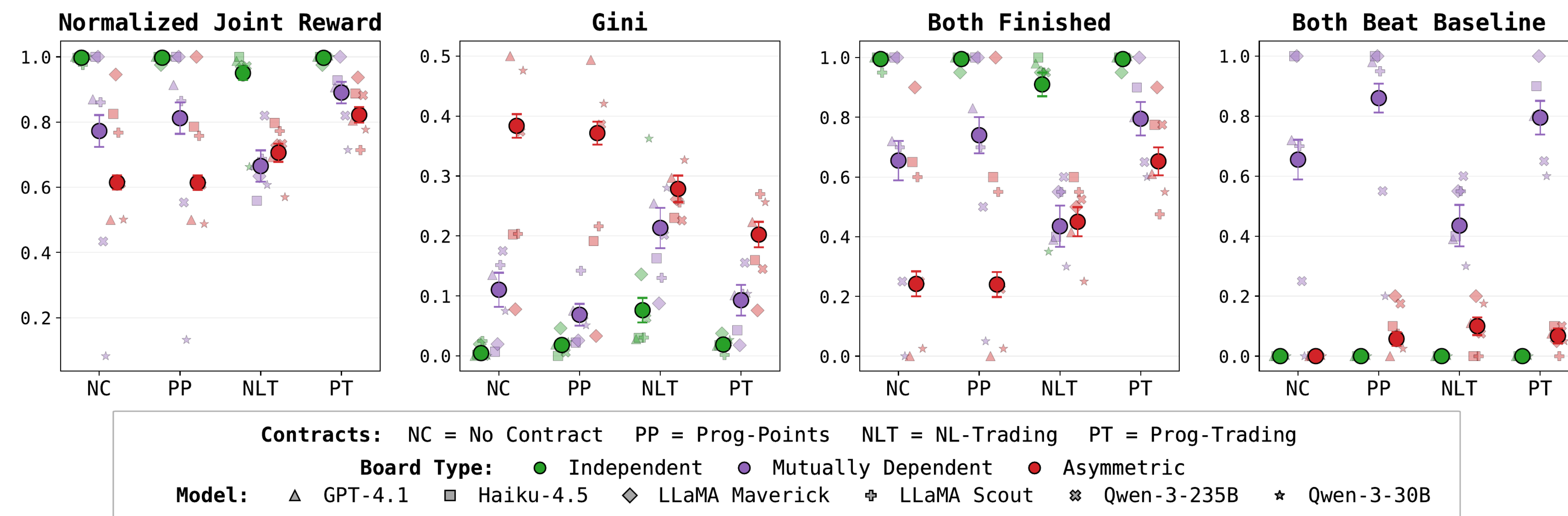
Agents negotiate their own contracts, in three forms:

- **Programmatic Trading**: chip transfers to cover agreed tiles, auto-executed
- **Natural Language Trading**: contract in natural language, an LLM judge enforces it
- **Programmatic Points**: promise part of your final score, auto-paid

A judge model (Qwen-3-235B) writes the contract; both players must accept it. Agents accept 86% of the time (Haiku-4.5: 67%). Programmatic Points ends winner-takes-all on 39% of Asymmetric boards (74% for Qwen models); Programmatic Trading assigns equal tiles in 49%.

Contracts Improve Cooperative Outcomes

Without contracts, no model ever got both players above their baseline on any Asymmetric board (BBB = 0.00 for all six models). With contracts:



Programmatic contracts improve cooperative outcomes.

Normalized Joint Reward rises from 0.77 (No-Contract) to 0.89 under Programmatic Trading ($p < 0.001$).

The gains are consistent across dependent boards.

Programmatic Trading has the highest Both Finished on 45 of the 60 Mutually Dependent and Asymmetric boards; No-Contract on 6.

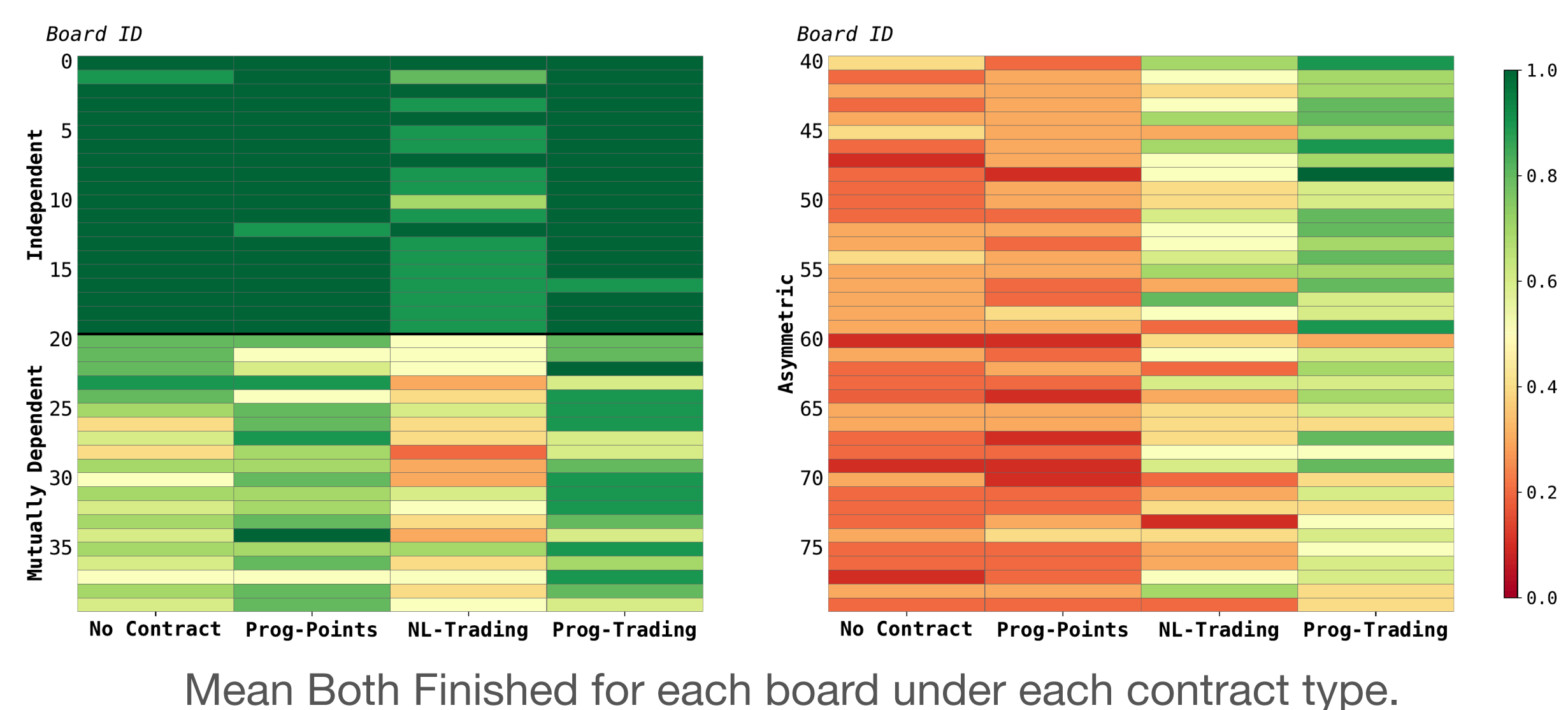
Enforced contracts reduce reliance on non-binding promises.

38% fewer Pay-for-Partner arrangements than without contracts.

On Asymmetric boards, the gains go to the weaker player.

P-Blue earns 68% more under Programmatic Trading ($p < 0.001$); **P-Red** does not gain.

Both Finished, Board by Board



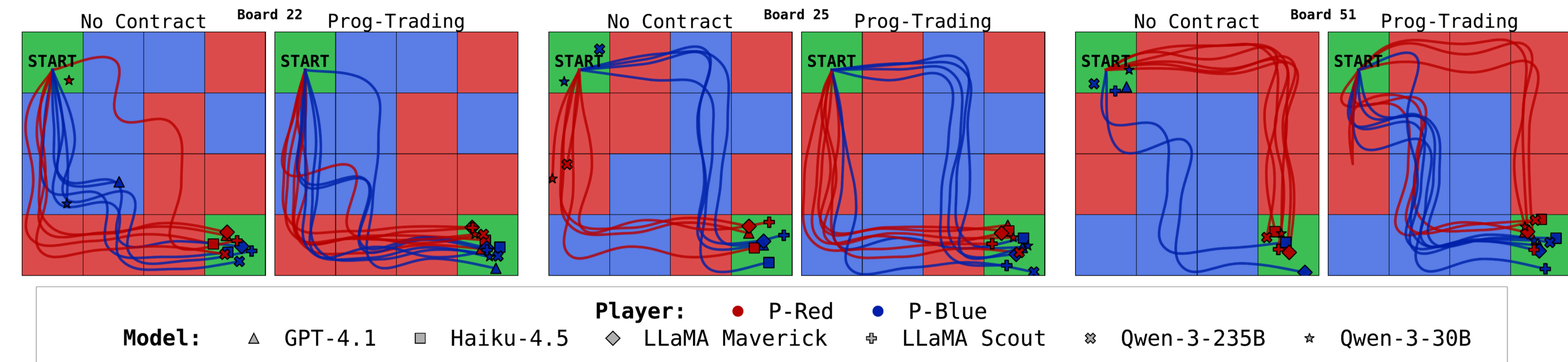
Mean Both Finished for each board under each contract type.

When Contracts Backfire

On Mutually Dependent boards, natural-language contracts do worse than no contract at all: both players finish 46% of the time, against 61% with no contract and 79% with Programmatic Trading.

The contracts formed are similar; the difference is downstream. With a natural-language contract, models propose fewer trades (1.4 vs 3.0 per game) and, when declining to trade, point to the contract 94% of the time (74% under Programmatic Trading). The contract anchors their reasoning and suppresses trading they still need.

Contracts Change Behavior on the Board



Contracts can improve outcomes. On boards where models fail to reach their goals without a contract, adding a Programmatic Trading contract lets every model complete its route.

Future Work & Extensions

A first step beyond self-play: pairing GPT-4.1 with Qwen-3-30B, which honors only 6% of its promises (Haiku-4.5 honors 100%), Programmatic Trading still enforced the transfers, and the strong model lifted the weak one at its own expense.

- Heterogeneous pairings and larger capability gaps
- Richer and formally verifiable contract mechanisms
- Moving beyond self-play throughout the benchmark

Takeaways

1. **The way contracts are represented and enforced matters**: contracts that compile to programs consistently outperform natural-language ones
2. **Self-negotiated contracts can improve cooperative outcomes** beyond regular trading
3. **Agents struggle to reach mutually beneficial outcomes from asymmetric starting points**

Paper & Acknowledgments



CT-Bench is open-sourced.

Paper: arxiv.org/abs/2607.22750

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